

Inas Bachiri

LinkedIn: [Inas Bachiri](#)

Github: [nousssss](#)

Google Scholar: [Inas Bachiri](#)

Email: inas.bachiri@nyu.edu / inas.bachiri@uphf.fr

Mobile: +33-745-318-257

Address: Valenciennes, France

EDUCATION

- **Doctor of Philosophy (PhD) in Computer Science** Valenciennes, France — Abu Dhabi, UAE
Université Polytechnique Hauts-de-France — INSA Hauts-de-France February 2024 - Present
 - **Thesis Title:** Software/Hardware Co-optimisation through Code Transformation and Hardware-aware Neural Architecture Search.
 - **Affiliation:** Joint project between New York University Abu Dhabi and the LAMIH laboratory at Université Polytechnique Hauts-de-France.
 - **Supervisors:** Prof. Smail Niar, Dr. Riyadh Baghdadi.
- **Engineering Degree in Intelligent Systems and Data - With High Honors** Algiers, Algeria
Higher National School of Computer Science (ESI ex INI) September 2019 - September 2024
 - **Main courses:** Machine Learning - Natural Language Processing - Combinatorial Optimization - Advanced Mathematics for Data Science - Complexity and Problem Solving - Computer Architecture - High Performance Computing - Distributed Computing and Artificial Intelligence - Data Mining and Analysis - Image Processing - Signal Processing - Knowledge Representation and Reasoning - Stochastic Processes, Models and Simulation - Linear Algebra - Multivariate Calculus - Probability and Statistics
 - **Average Grade:** 14.48/20
- **Master Degree in Intelligent Systems and Data Science - With High Honors** Algiers, Algeria
Higher National School of Computer Science (ESI ex INI) September 2023 - September 2024
 - **Main courses:** Introduction to research methodology - Research Communication and Literature Review - Information Retrieval and Access - Information Extraction
 - **Average Grade:** 16.35/20

EXPERIENCE

- **Doctoral Researcher** February 2024 - Present
LAMIH UMR CNRS 8201 - Université Polytechnique Hauts-de-France Valenciennes, France
 - **Thesis Project:** Currently pursuing doctoral research on the efficient deployment of AI models on edge devices. [My thesis](#) is a collaborative project between New York University Abu Dhabi and the LAMIH laboratory at Université Polytechnique Hauts-de-France, supported by a scholarship from INSA Hauts-de-France.
- **Research Intern** September 2023 - August 2024
New York University Abu Dhabi (NYUAD) Abu Dhabi, UAE (Remote)
LAMIH UMR CNRS 8201 - Université Polytechnique Hauts-de-France Valenciennes, France (On Site)
 - **Internship Project:** Reviewing literature on the combination of Neural Architecture Search and Compiler Optimization for neural network acceleration ([Manuscript](#) - [Preprint Survey](#)), then designing and implementing a Hardware-Aware Neural Architecture Search framework that allows the integration of Automatic Code Optimization in the MLIR Compiler (See the **CONAS** framework in the Projects section).
- **Process Mining Intern** September 2022 - October 2022
Centre de Développement des Technologies Avancées (CDTA) Algiers, Algeria
 - **Internship Project:** Design and Implementation of a ProM (Process Mining Framework) extension to analyze and solve interoperability issues between processes (See **Interoperability Solver** in the Projects section).

PUBLICATIONS

- **Neural Architecture Search and Automatic Code Optimization: Techniques, Trends, and Challenges**
Journal of Systems Architecture (Elsevier, Q1)
 - **Paper Reference:** Bachiri, I., Ouarnoughi, H., Niar, S., & Baghdadi, R. (2025). *Neural Architecture Search and Automatic Code Optimization: Techniques, trends, and challenges*. Journal of Systems Architecture, 103645. <https://doi.org/10.1016/j.sysarc.2025.103645>

TECHNICAL SKILLS

- **Programming Languages:** C/C++ - Java - Python - Julia - Bash scripting - R - Shell Scripting
- **Data Science and AI Frameworks:** PyTorch - Tensorflow and Keras - Scikit-learn - OpenCV - pandas - numpy - matplotlib
- **Databases:** MySQL - MongoDB - Oracle
- **High Performance Computing:** CUDA - OpenMP - MPI - Pthreads - Python multithreading

PROJECTS

- **CONAS: Compiler Optimization with Neural Architecture Search**

LAMIH CNRS & NYUAD

September 2023 - August 2024

- **Description:** CONAS is a Hardware-Aware Neural Architecture Search framework that integrates Automatic Code Optimization through the MLIR compiler. The framework accelerates convolutional neural networks by discovering novel, efficient operators that replace traditional convolution layers, maintaining performance and enhancing latency on resource-constrained devices.
- **Links:** [Manuscript](#) - [Code](#)
- **Keywords:** Efficient Deep Learning, AutoML, Compiler Optimization, MLIR.

- **AI Explainability in Medical Imaging: A Case Study on Breast Cancer Detection**

Research Summer Camp - School of AI Algiers

July 2023 - October 2023

- **Description:** Analyzing and benchmarking mathematical methods to explain and interpret deep learning models for breast cancer detection.
- **Links:** [Poster](#) - [Code](#)
- **Keywords:** Deep Learning, AI Explainability, Mammography, Applied Mathematics.

- **Flowshop Scheduling Optimization with AI**

Higher National School of Computer Science

February 2023 - May 2023

- **Description:** Research project focused on improving a metaheuristic using machine learning techniques to solve the Flowshop scheduling problem.
- **Links:** [Manuscript](#)
- **Keywords:** Reinforcement Learning, Combinatorial Optimization, Hybridization.

- **Community Detection in Social Networks**

Higher National School of Computer Science

February 2023 - May 2023

- **Description:** Research project focusing on the fusion of combinatorial optimization and machine learning techniques to solve the community detection problem.
- **Links:** [Manuscript](#) - [Code](#)
- **Keywords:** Machine Learning, Combinatorial Optimization, Graphs, Complex Networks.

- **Interoperability Solver**

Centre de Développement des Technologies Avancées (CDTA)

September 2022 - October 2022

- **Description:** Interoperability Solver is a process mining plugin for ProM that helps solve interoperability problems. The plugin enables users to verify the conformity between an event log and its associated process model, repair the model based on the result of the alignment, and annotate the activities of the repaired model according to the content and type of messages associated with the events.
- **Links:** [Code](#)
- **Keywords:** Graphs, Optimizing Data Structures, Advanced OOP, Multi-threaded Programming

LANGUAGE PROFICIENCY

- **English Language Proficiency (C2) – EF SET**

EF Standard English Test

November 2023

- **Link:** [Language Proficiency Certificate](#)

- **French Language Proficiency (C2) – TCF SO (Test de Connaissance du Français)**

Institut français d'Algérie

May 2023

- **Link:** [Language Proficiency Certificate](#)

VOLUNTEERING

- **Reviewer at AAAI** August 2025
 - Reviewed a submission to the [AAAI](#) (Association for the Advancement of Artificial Intelligence) 2026 Conference.
- **Reviewer at IJCNN** February 2025
 - Reviewed a submission to the [IJCNN](#) (International Joint Conference on Neural Networks) 2025 Conference.
- **Reviewer at DAC** January 2025
 - Reviewed two submissions to the [DAC](#) (Design Automation Conference) 2025 Conference.
- **Reviewer at DATE** November 2024
 - Reviewed a submission to the [DATE](#) (Design, Automation and Test in Europe) 2025 Conference.
- **Guest Lecturer at Batam State Polytechnic** November 2024
 - Presented an online lecture for the Data Mining course at Batam State Polytechnic in Indonesia, guiding students on how to approach research projects in AI for their final-year projects.
- **Reviewer MINT @ NEURIPS** September 2024
 - Reviewed a submission to the [MINT Workshop](#) (Workshop on Foundation Model Interventions) at the NeurIPS 2024 Conference.
- **Notion Campus Leader** September 2023 - August 2024
 - Represented Notion at my university by organizing workshops and tutoring sessions. Created resources to support students and collaborated with other Campus Leaders to promote Notion's value within academic communities.
- **Reviewer at DSD** June 2024
 - Reviewed a submission to the [DSD](#) (Digital System Design) 2024 Conference.
- **Workshop Presenter at School of AI Algiers** December 2023
 - Prepared learning materials and assignments, and presented workshops at [AI Camp](#) (an AI bootcamp) namely: Maths For AI, Image classification and Transfer Learning.
- **Reviewer at NuClear @ AAAI 2024** December 2023
 - Reviewed two submissions to the [NuClear Workshop](#) (Neuro-Symbolic Learning and Reasoning in the Era of Large Language Models) at the AAAI 2024 Conference.
- **Cybersecurity Awareness Presenter at Breaking Science (Charity Science Exhibition)** March 2020
 - Represented the Shellmates Club and demonstrated social engineering attacks and provided practical tips to help individuals protect themselves online.

CERTIFICATIONS

- **Participation in ARCHI Spring School**
Laboratoire d'informatique, de robotique et de microélectronique de Montpellier December 2024
 - **Overview:** Attended the [ARCHI](#) Spring School, about embedded hardware/software system architectures.
 - **Link:** [Participation Certificate](#)
- **Participation in AI Summit**
School of AI Algiers December 2023
 - **Overview:** Presented a research poster about Explainability in Medical Imaging at the [AI Summit](#) event.
 - **Link:** [Participation Certificate](#)
- **NeuroSymbolic AI Summer School**
2nd Neuro-Symbolic Summer School 2023 August 2023
 - **Overview:** Attended the [IBM NSSS 2023](#), which include talks, panels, and tutorials in various areas of theory and the application of neuro-symbolic AI
 - **Link:** [Completion Certificate](#)
- **Participation in IWD Hackathon**
Women Techmakers Algiers March 2023
 - **Overview:** Participated in the IWD'23 Hackathon by working on the [BeeFriendly app](#). Contributed within my team by implementing a computer vision model to classify bee hives based on their health status.
 - **Link:** [Participation Certificate](#)